BTS-II(R)-07-24-2824 Reg. No.

A

# B.Tech. Degree II Semester Regular Examination July 2024

## EE/EC/CS/IT 23-200-0201B LINEAR ALGEBRA AND TRANSFORM TECHNIQUES

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve linear system of equations and to determine Eigen values and vectors of a matrix.
CO2: Exemplify the concept of vector space and subspace.
CO3: Determine Fourier series expansion of functions and transform.
CO4: Solve linear differential equation and integral equation using Laplace transform.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

### PART A

(Answer *ALL* questions)

(5 × 2 = 10)

| | | | Marks | BL | CO | PO |
|---|---|---|---|---|---|---|
| I. | (a) | Find the characteristic equation of $A=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$. | 2 | L1 | 1 | 2 |
| | (b) | Show that {(1,0), (0,1)} forms a basis for the vector space $R^2$ over R. | 2 | L1 | 2 | 2 |
| | (c) | Is the set of all real numbers R over the set of all complex numbers C, a vector space? Justify your answer. | 2 | L2 | 2 | 2 |
| | (d) | Show that $F_s\{xf(x)\}=-\frac{d}{ds}F_c\{f(x)\}$. | 2 | L1 | 3 | 2 |
| | (e) | Find $L\{t\sin 5t\}$. | 2 | L2 | 4 | 2 |

### PART B

(4 × 10 = 40)

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|---|---|---|---|---|---|---|
| II. | (a) | Solve the system of equations $2x+3y+4z=31,\ x-2y+4z=4,\ 3x+y-3z=2,\ 6x+2y-6z=4$. | 5 | L3 | 1 | 2 |
| | (b) | Find the eigen values and eigen vectors of $A=\begin{bmatrix}1 & 0 & -1\\ 1 & 2 & 1\\ 2 & 2 & 3\end{bmatrix}$. | 5 | L4 | 1 | 2 |
| | | **OR** | | | | |
| III. | (a) | Find $l$ and $m$ if the rank of the matrix $A=\begin{pmatrix}0 & 1 & -3 & -1\\ 1 & 0 & 1 & 1\\ 3 & 1 & l & m\\ 1 & 1 & -2 & l\end{pmatrix}$ is 2. | 5 | L5 | 1 | 2 |
| | (b) | Using Cayley Hamilton theorem find the inverse of the matrix. $A=\begin{bmatrix}1 & 0 & 3\\ 2 & 1 & -1\\ 1 & -1 & 1\end{bmatrix}$ | 5 | L4 | 1 | 2 |

| | | | Marks | BL | CO | PO |
|---|---|---|---|---|---|---|
| IV. | (a) | Determine whether the vectors (1, 3, 2), (1, -7, 8) and (2, 11, -1) are linearly independent. | 5 | L3 | 2 | 2 |
| | (b) | Let V be the vector space of set of all $2 \times 2$ matrices. Prove that dim (V) = 4. | 5 | L4 | 2 | 2 |
| | | **OR** | | | | |
| V. | (a) | Find an orthonormal basis for the subspace W of the vector space $R^3$ over the field R generated by the vectors (2, 1, 1) and (1, 3, -1). | 5 | L5 | 2 | 2 |
| | (b) | Let V and W be two vector spaces over the same field F and let T be a linear transformation from V to W, prove that Ker T is a subspace of V and Im T is a subspace of W. | 5 | L4 | 2 | 2 |
| VI. | (a) | Find the Fourier series expansion for $f(x)=\begin{cases} x+x^2, & -\pi<x<\pi \\ \pi^2, & x=\pm\pi \end{cases}$. Hence prove that $\frac{\pi^2}{6}=1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\ldots$ | 5 | L3 | 3 | 2 |
| | (b) | Find the Fourier integral representation of $f(x)=\begin{cases} 1, & \lvert x\rvert<1 \\ 0, & \lvert x\rvert>1 \end{cases}$. Hence prove that $\int_0^\infty \frac{\cos x \sin x}{x}\,dx = \frac{\pi}{4}$. | 5 | L6 | 3 | 2 |
| | | **OR** | | | | |
| VII. | (a) | Find the Fourier transform of $e^{-a^2x^2}$, $a>0$, hence deduce that $e^{\frac{-x^2}{2}}$ is self reciprocal with respect to Fourier transform. | 5 | L5 | 3 | 2 |
| | (b) | Find the Fourier sine transform of $e^{-\lvert x\rvert}$. Hence evaluate $\int_0^\infty \frac{x\sin mx}{1+x^2}\,dx$. | 5 | L6 | 3 | 2 |
| VIII. | (a) | Evaluate $\int_0^\infty \frac{e^{-4t}\sin^2 t}{t}\,dt$ using Laplace transform. | 5 | L5 | 4 | 2 |
| | (b) | Find the inverse Laplace transform of $\frac{5s^2-15s-11}{(s+1)(s-2)^3}$. | 5 | L3 | 4 | 2 |
| | | **OR** | | | | |
| IX. | (a) | State Convolution theorem and evaluate $L^{-1}\left\{\frac{1}{(s^2+1)(s^2+9)}\right\}$. | 5 | L4 | 4 | 2 |
| | (b) | Solve $y''-3y'+2y=e^{2x}$, $y(0)=-3$, $y'(0)=5$. | 5 | L6 | 4 | 2 |

Bloom's Taxonomy Levels
L1 – 14.3%, L2 – 9.5%, L3 – 19%, L4 – 23.9%, L5 – 19%, L6 – 14.3%.

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